NED University of Engineering & Technology

Department of Computer Science & Information Technology

Web Engineering | CT-353

Click & Cook

Smart Kitchen Companion

An AI-Powered Personalized Recipe Recommender

Project Documentation Report

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Live Demo: https://click-and-cook-123.vercel.app/

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1. Project Overview

Click & Cook is a full-stack AI-powered web application that transforms the cooking experience by generating personalized recipes based on user-provided ingredients, dietary preferences, and calorie targets. The application leverages Google's Gemini 2.5 Flash large language model to craft unique, high-quality recipes in real-time, fetches appetizing food images via Unsplash API, and maintains a smart library of saved recipes for each authenticated user.

The platform bridges the gap between what users have in their pantry and what they can cook, eliminating food waste and inspiring culinary creativity. By analyzing saved recipe history, the Personalized Picks feature intelligently recommends new dishes tailored to each user's unique taste profile.

Application Name	Click & Cook
Type	Full-Stack Web Application
Category	AI-Powered Recipe Recommender
Target Users	Home cooks, food enthusiasts, health-conscious individuals
Live URL	https://click-and-cook-123.vercel.app/
Backend URL	https://click-and-cook.onrender.com
GitHub Repository	https://github.com/eishasohail/Click-and-Cook-/

2. Problem Statement

Modern home cooks face a recurring challenge: they have various ingredients available but lack inspiration or knowledge to create satisfying meals. This leads to:

- Food waste due to unused or expiring ingredients
- Over-reliance on repetitive meal patterns
- Time wasted searching through recipe websites that don't match available ingredients
- Difficulty maintaining dietary goals and calorie targets
- Lack of personalization in generic recipe recommendations

Click & Cook directly addresses these pain points by allowing users to input exactly what they have, specify their calorie requirements, and receive an AI-generated recipe perfectly tailored to their situation — instantly and intelligently.

3. Objectives

- Develop a production-ready full-stack web application using modern technologies
- Integrate Google Gemini 2.5 Flash AI for intelligent recipe generation
- Implement secure JWT-based authentication with Google OAuth support
- Build a Smart Library for organizing and managing saved recipes by category
- Create an AI-driven Personalized Picks system based on user taste profiles
- Fetch relevant food images automatically via Unsplash API
- Deploy the application on cloud platforms (Vercel + Render) for public access
- Ensure a premium, responsive UI/UX consistent with modern design standards

4. Tech Stack

4.1 Frontend

Technology	Version	Purpose
React.js	18.x	UI framework and component architecture
Tailwind CSS	3.x	Utility-first CSS styling
Framer Motion	10.x	Animations and page transitions
Redux Toolkit	2.x	Global state management for auth
React Router	6.x	Client-side routing
Lucide React	Latest	Icon library
Axios	1.x	HTTP client for API calls

4.2 Backend

Technology	Version	Purpose
Node.js	22.x	JavaScript runtime environment
Express.js	4.x	REST API framework
PostgreSQL	15.x	Relational database
Neon Cloud	Cloud	Serverless PostgreSQL hosting
bcryptjs	2.x	Password hashing
cors	2.x	Cross-origin resource sharing
jsonwebtoken	9.x	JWT authentication tokens

4.3 External APIs & Services

Service	Plan	Usage
Google Gemini 2.5 Flash	Free Tier	AI recipe generation & personalization
Unsplash API	Free Tier	Automatic food photography fetching
Google OAuth 2.0	Free	Social login authentication
Vercel	Free	Frontend deployment & hosting
Render	Free	Backend deployment & hosting

5. System Architecture

Click & Cook follows a modern three-tier architecture with clear separation between the presentation layer, business logic layer, and data layer.

5.1 Architecture Overview

Layer	Components & Technologies
Presentation Layer	React.js + Tailwind CSS + Framer Motion — deployed on Vercel
Business Logic Layer	Node.js + Express.js REST API — deployed on Render
Data Layer	PostgreSQL on Neon Cloud — serverless database
AI Layer	Google Gemini 2.5 Flash API — recipe intelligence
Media Layer	Unsplash API — automatic food photography
Auth Layer	JWT Tokens + Google OAuth 2.0

5.2 Data Flow

- User authenticates via JWT or Google OAuth
- Frontend (React) makes API calls to Express backend on Render
- Backend processes requests, queries PostgreSQL on Neon, calls Gemini AI
- Gemini returns structured JSON recipe data
- Backend fetches food image from Unsplash using recipe name as search query
- Complete recipe with image URL is returned to frontend
- User can save recipe — stored in PostgreSQL with all fields
- Smart Library fetches saved recipes for display with filtering/search
- Personalized Picks analyzes saved recipes and queries Gemini for suggestions

5.3 State Management

The application uses Redux Toolkit for global state management with the following slices:

- authSlice: Manages user authentication state (user object, JWT token, isAuthenticated flag)
- State persists across page refreshes via localStorage synchronization
- All API calls use token from Redux state for authorization headers

6. Database Schema

The application uses PostgreSQL hosted on Neon Cloud with three main tables:

6.1 Users Table

Column	Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Unique user identifier
first_name	VARCHAR(100)	NOT NULL	User's first name
last_name	VARCHAR(100)		User's last name
email	VARCHAR(255)	UNIQUE NOT NULL	User's email address
password_hash	TEXT	NOT NULL	Bcrypt hashed password
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

6.2 Saved Recipes Table

Column	Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Recipe record identifier
user_id	INTEGER	FOREIGN KEY	References users(id)
title	VARCHAR(255)	NOT NULL	AI-generated recipe name
ingredients	TEXT		Formatted ingredients list
instructions	TEXT		Full recipe instructions
calories	INTEGER		Calculated calorie count
servings	INTEGER		Number of servings
image_url	VARCHAR(500)		Unsplash food image URL
category	VARCHAR(50)	DEFAULT 'Dinner'	Recipe category classification
prep_time	INTEGER		Preparation time in minutes

cook_time	INTEGER		Cooking time in minutes
difficulty	VARCHAR(20)		Easy / Medium / Hard
nutritional_info	JSONB		Protein, carbs, fats data
created_at	TIMESTAMP	DEFAULT NOW()	Save timestamp

6.3 Reviews Table

Column	Type	Constraint	Description
id	SERIAL	PRIMARY KEY	Review identifier
user_id	INTEGER	FOREIGN KEY	References users(id)
name	VARCHAR(100)		Reviewer display name
text	TEXT	NOT NULL	Review content
rating	INTEGER	DEFAULT 5	Star rating 1-5
accent_color	VARCHAR(10)		Display color for review card
created_at	TIMESTAMP	DEFAULT NOW()	Review submission time

7. API Routes

All protected routes require a valid JWT token in the Authorization header: Bearer <token>

7.1 Authentication Routes

Method	Endpoint	Description
POST	/Signup	Register new user, returns JWT token + user object
POST	/login	Authenticate user, returns JWT token + user object
POST	/api/auth/google	Google OAuth login with access token or ID token

7.2 Recipe Routes

Method	Endpoint	Description
POST	/api/recipes/generate	Generate recipe via Gemini AI + fetch Unsplash image [Protected]
POST	/api/recipes/follow-up	Answer follow-up questions about generated recipe [Protected]
POST	/api/recipes/save	Save recipe to user's Smart Library [Protected]
GET	/api/recipes/saved	Fetch all saved recipes for authenticated user [Protected]
DELETE	/api/recipes/:id	Delete a specific saved recipe by ID [Protected]
GET	/api/recipes/recommendations	Get AI-powered personalized picks + taste profile [Protected]

7.3 Reviews Routes

Method	Endpoint	Description
GET	/api/reviews	Fetch latest 3 reviews for landing page display
POST	/api/reviews	Submit a new review with rating [Protected]

8. Features & Screens

8.1 Landing Page (/)

The landing page serves as the public-facing entry point for Click & Cook. It features:

- Full-screen hero section with animated food background imagery
- Feature showcase cards: AI Recipe Generation, Smart Library, Personalized Picks, Save Your Favorites
- User testimonials/reviews section fetched dynamically from the database
- Navigation to Sign In and Sign Up flows
- Smooth scroll animations and hover effects throughout

8.2 Authentication Screens (/signin, /signup)

Premium glassmorphism design with full-screen background food imagery:

- Two-column layout: left branding panel + right glass card form
- Sign In: email/password fields with show/hide toggle, Google OAuth button, JWT token management
- Sign Up: first name, last name, email, password, confirm password with full validation
- Redux state management synced with localStorage for session persistence
- Animated error messages with shake effect, success state with ChefHat animation
- Toggle between Login/Signup tabs with smooth transitions

8.3 Dashboard (/dashboard)

Personalized chef's workspace with dynamic content:

- Hero section with full-bleed background image and personalized welcome message
- Shimmer text effect on user's name, transparent stat cards (recipes saved, generated)
- Three feature cards with liquid-fill button animations: Recipe Generator, My CookBook, Personalized Picks
- Review submission form with interactive star rating (1-5 stars), real-time form validation
- Submitted reviews appear on the landing page for community visibility

8.4 AI Recipe Generator (/ai-recipes)

The core feature of the application — a premium AI-powered recipe generation interface:

- Left sidebar (Recipe Architect): ingredient tags management, calorie range slider (100-2000 kcal), servings counter, dietary notes
- Right panel (AI Chat Interface): displays conversation with Chef AI assistant
- Gemini 2.5 Flash generates structured JSON recipes with name, category, ingredients, instructions, nutritional info, prep/cook time, difficulty
- Unsplash API automatically fetches a relevant food image using the recipe name as search query
- Recipe card displays: food image, title, time/calories/servings stats, key ingredients grid, numbered instructions
- Follow-up chat: users can ask questions about the recipe (substitutions, spice levels, storage tips)
- Save Recipe button saves to Smart Library with all metadata

- Backup API key fallback — automatically switches to secondary key if primary quota is exhausted

8.5 My CookBook (/library)

An organized digital cookbook for managing all saved recipes:

- Recipe cards grid (3 columns desktop) showing food image, category badge, calories, servings, date saved
- 7 category filter pills: Breakfast, Lunch, Dinner, Desserts, Drinks, Healthy, Snacks
- Real-time search by recipe name
- Recipe count badge dynamically updating
- Delete recipe with confirmation and smooth exit animation
- View Details modal: tabs for Ingredients and Steps, nutritional info (protein/carbs/fats), stats grid (time/difficulty/servings/calories)
- Empty state with animated icon when no recipes saved
- Skeleton loading cards while fetching

8.6 Personalized Picks (/personalized-picks)

AI-powered recommendation engine analyzing user taste preferences:

Personalized Picks is an intelligent recommendation engine that learns from user behavior and generates tailored recipe suggestions.

- Taste Profile Analysis: System analyzes all saved recipes to identify patterns
- Favorite Category Detection: Determines most frequently saved recipe category
- Ingredient Frequency Analysis: Identifies most commonly used ingredients
- Intelligent Tie-Breaking: When categories are equal, prioritizes most recently saved recipe
- AI Generation: Feeds taste profile data to Gemini to generate 3 personalized suggestions
- Performance Caching: 30-minute cache for fast performance without excessive API calls
- Refresh on Demand: Users can generate fresh recommendations anytime

9. AI Integration

9.1 Google Gemini 2.5 Flash

The application uses two Gemini model configurations:

JSON Mode (Recipe Generation)

When generating recipes, Gemini is configured with responseMimeType: 'application/json' to return structured data. The system instruction defines a strict schema:

- recipeName: Creative, descriptive name
- category: Classified as Breakfast/Lunch/Dinner/Desserts/Drinks/Healthy/Snacks
- prepTime, cookTime: Time in minutes
- calories, servings: Nutritional quantities
- ingredients: Array of objects with item, amount, unit
- instructions: Array of objects with instruction text and duration
- nutritionalInfo: Object with protein, carbs, fat values
- imageSearchQuery: 3-4 word Unsplash search term

Text Mode (Follow-up Questions)

For conversational follow-up questions, Gemini uses plain text mode with context from the generated recipe, responding concisely as a professional chef.

9.2 Category Intelligence

Since Gemini occasionally omits the category field from responses, a robust fallback system analyzes:

- dietaryTags array: checks for 'breakfast', 'brunch', 'vegan', 'healthy' etc.
- Recipe name keywords: 'smoothie' → Drinks, 'cake' → Desserts, 'biryani' → Dinner
- Calorie count: recipes under 300 calories default to Healthy category
- Default fallback: Dinner (covers most unrecognized main dishes)

9.3 Rate Limit Management

The free tier of Gemini 2.5 Flash provides 20 requests/day per API key. To ensure uninterrupted service, Click & Cook implements sophisticated rate limit handling.

Multi-Key Architecture:

- **Primary Key:** GEMINI_API_KEY (Main production key)
- **Backup Keys:** GEMINI_API_KEY_BACKUP, GEMINI_API_KEY_BACKUP_2, GEMINI_API_KEY_BACKUP_3

Automatic Fallback Logic:
When the primary key receives a 429 (Too Many Requests) error, the system automatically switches to the next available key without user disruption. This allows the application to serve up to 80 requests daily (4 keys × 20 requests). **JSON Repair Mechanism:**
To prevent response truncation, maxOutputTokens is set to 8192. A JSON repair utility strips

markdown wrappers and fixes trailing commas before parsing to ensure robustness.

9.4 Unsplash Image Integration

After Gemini generates a recipe, the backend automatically fetches high-quality food photography:
Image Fetching Process:

1. Extracts imageSearchQuery field from recipe JSON (e.g., "spicy chicken curry")
2. Calls Unsplash API: /search/photos?query={term}&per;_page=1
3. Retrieves and saves the high-quality food image URL to database
4. Falls back to default image if Unsplash returns no results
5. Images stored as URLs (not binary data) for efficient database usage

Similar to Gemini, multiple Unsplash API keys are configured (3 keys) to handle rate limits gracefully.

10. Authentication

10.1 JWT Authentication Flow

- User submits credentials → backend validates → signs JWT with 24h expiry
- Token stored in Redux state + localStorage for persistence across sessions
- All protected API routes check Authorization: Bearer <token> header
- Token decoded using jsonwebtoken to extract userId and email
- Passwords hashed with bcryptjs (10 salt rounds) — never stored in plaintext

10.2 Google OAuth Flow

- Frontend uses @react-oauth/google library with useGoogleLogin hook
- User clicks Google button → Google popup → returns access token
- Frontend sends access token to /api/auth/google endpoint
- Backend fetches user info from Google userinfo endpoint
- If user exists: returns JWT. If new: creates account then returns JWT
- OAuth users stored with placeholder password hash (no password)

10.3 Redux Auth Slice

The authSlice manages the following state:

- user: Object containing firstName and email
- token: JWT string for API authentication
- isAuthenticated: Boolean flag for route protection
- setCredentials action: stores user+token in Redux and localStorage
- logout action: clears Redux state and localStorage simultaneously

11. Deployment

11.1 Frontend — Vercel

Platform	Vercel
Build Command	npm run build
Root Directory	client/
Framework	Create React App
Environment Variable	VITE_API_BASE_URL=http://localhost:5000 VITE_GOOGLE_CLIENT_ID=147393390596-i6vn54sv0j7renvmm8qdni76dsgk3mvu.apps.googleusercontent.com

Live URL	https://click-and-cook-123.vercel.app/

11.2 Backend — Render

Platform	Render Web Service
Root Directory	server/
Build Command	npm install
Start Command	node app.js
Environment Variables	DATABASE_URL=postgresql://neondb_owner:npg_lDy9GkHaEjd3@ep-cool-cake-ams5u1gl.c-5.us-east-1.aws.neon.tech/neondb?sslmode=require GEMINI_API_KEY=AIzaSyB-DVNzsZLE7pVUXSBSxJ9pKUdCm4cl49c GEMINI_API_KEY_BACKUP=AIzaSyC_NRceM6Pjpzi72S8y9XeNu90hGvNYfVQ GEMINI_API_KEY_BACKUP_2=AIzaSyCLzoIUxKuCRlr-z8TxfP_dfms9PhW378I GEMINI_API_KEY_BACKUP_3=AIzaSyA54brXYDT0l4fSxt8awzNv-tkxsPyl0ps JWT_SECRET=recipe123secret PORT=5000 GOOGLE_CLIENT_ID=147393390596-i6vn54sv0j7renvmm8qdni76dsgk3mvu.apps.googleusercontent.com GOOGLE_CLIENT_SECRET=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 GOOGLE_CALLBACK_URL=http://localhost:5000/auth/google/callback UNSPLASH_ACCESS_KEY=HLXM-THSVejjm3TuIrBFZlw3czdIm6ptaZZthEm3TiE UNSPLASH_ACCESS_KEY_2=Dky_TOOIBN8Gjr4eD8ly-xGaBegdefXwflX-T3bP84U UNSPLASH_ACCESS_KEY_3=REPLACE_WITH_THIRD_KEY
Backend URL	https://click-and-cook.onrender.com

11.3 Database — Neon Cloud

Platform	Neon Serverless PostgreSQL
Project Name	recipe-recommender-app
Database	neondb
Tables	users, saved_recipes, reviews
Connection	SSL-enabled connection string via DATABASE_URL env variable

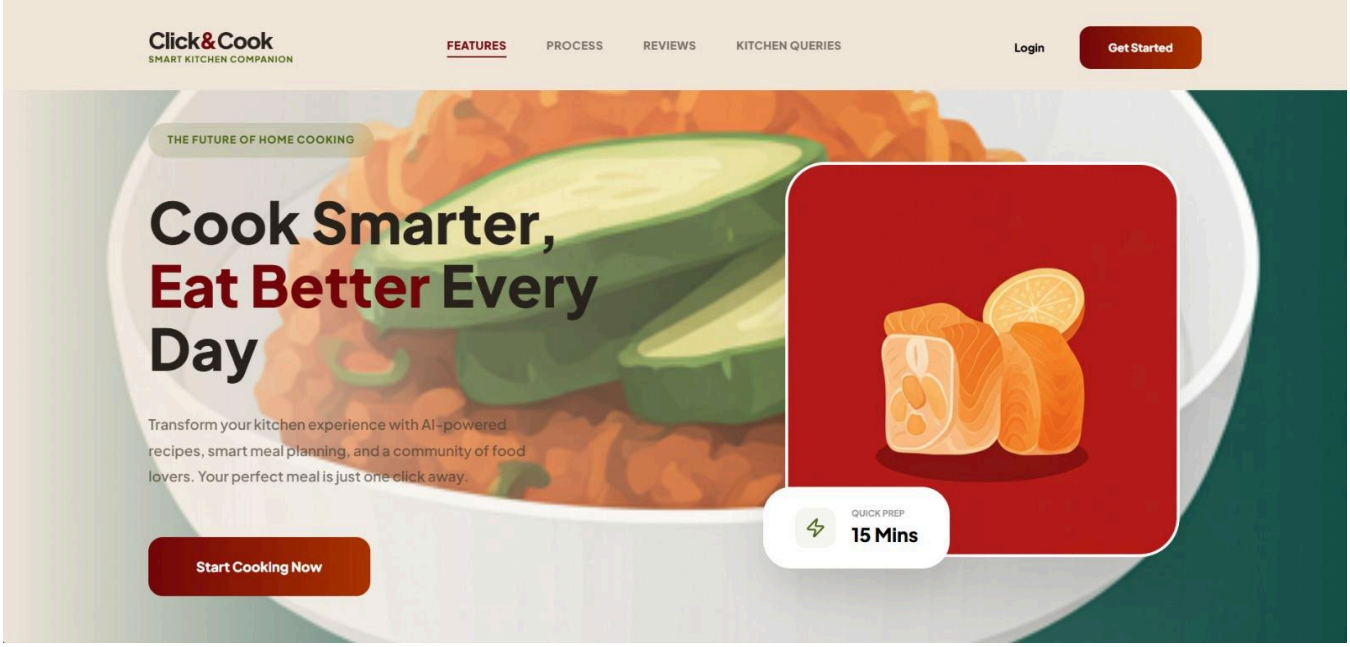
12. Challenges & Solutions

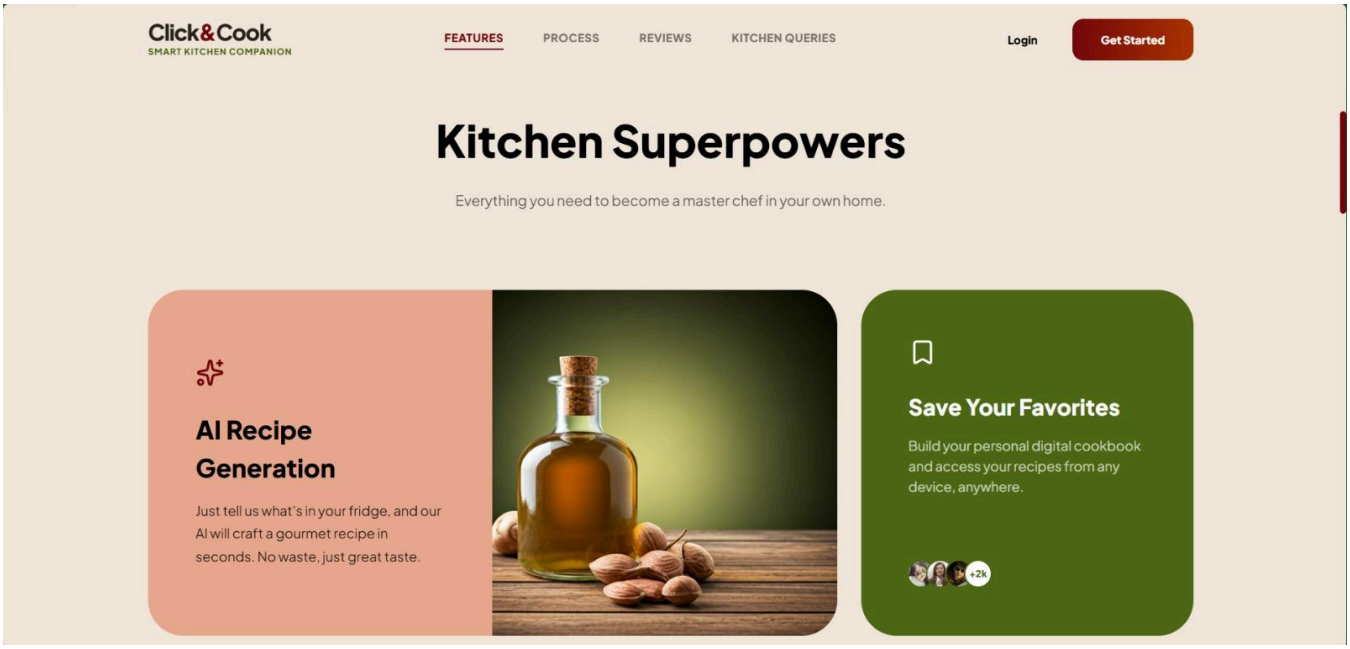
Challenge	Solution Implemented
Gemini JSON Truncation	Increased maxOutputTokens from 4096 to 8192. Added JSON repair logic to fix truncated responses before parsing.
Category Classification	Gemini often omitted the category field. Implemented fallback system analyzing dietaryTags array and recipe name keywords with comprehensive keyword lists.
API Rate Limits	Implemented API key support with automatic fallback. Primary key exhausted → seamlessly switches to backup kes without user disruption.
Image Relevance	Gemini generates a specific imageSearchQuery field optimized for Unsplash rather than using the full recipe name, improving image relevance.
Auth State Persistence	Redux authSlice initializes from localStorage on app load. setCredentials and logout actions keep Redux state and localStorage perfectly synchronized.
Google OAuth Integration	Used @react-oauth/google with useGoogleLogin hook. Backend verifies access token via Google userinfo endpoint rather than ID token for reliability.
CORS Configuration	Configured Express CORS middleware to allow requests from both localhost:3000 (development) and Vercel production URL.
Database Schema Evolution	Used ALTER TABLE IF NOT EXISTS for all new columns. Created migrate.js script for running schema updates without data loss.

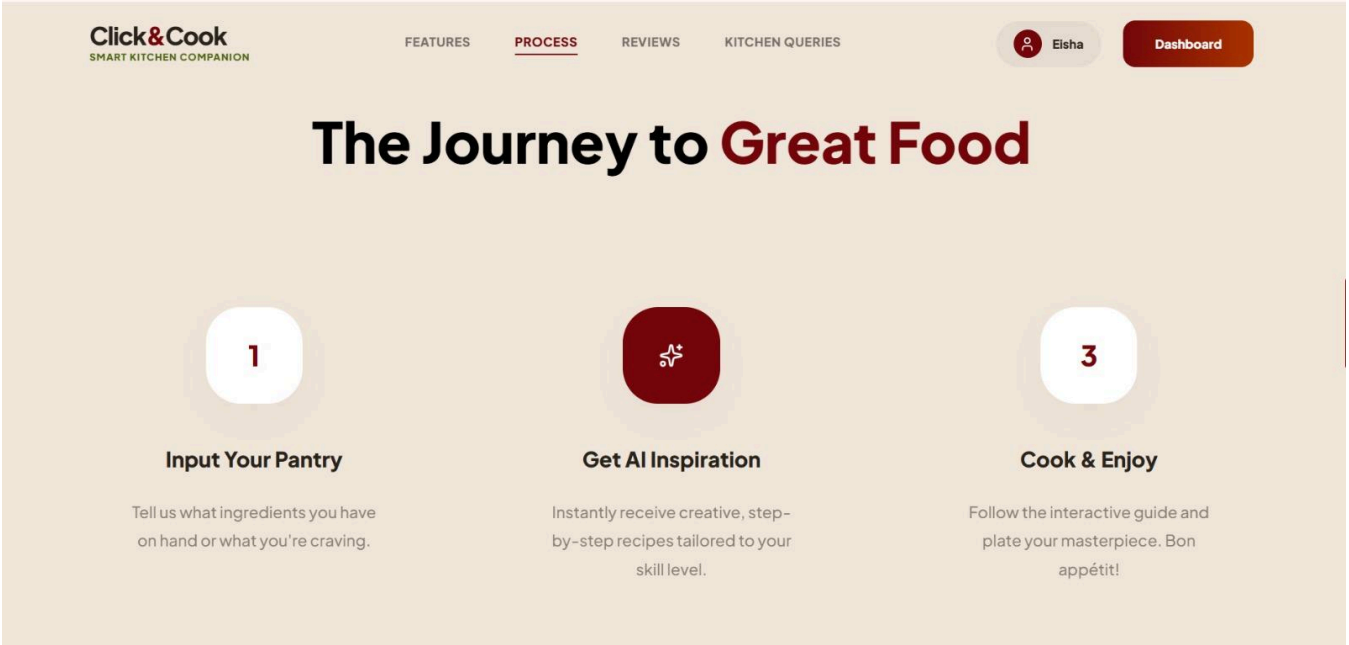
13. Future Enhancements

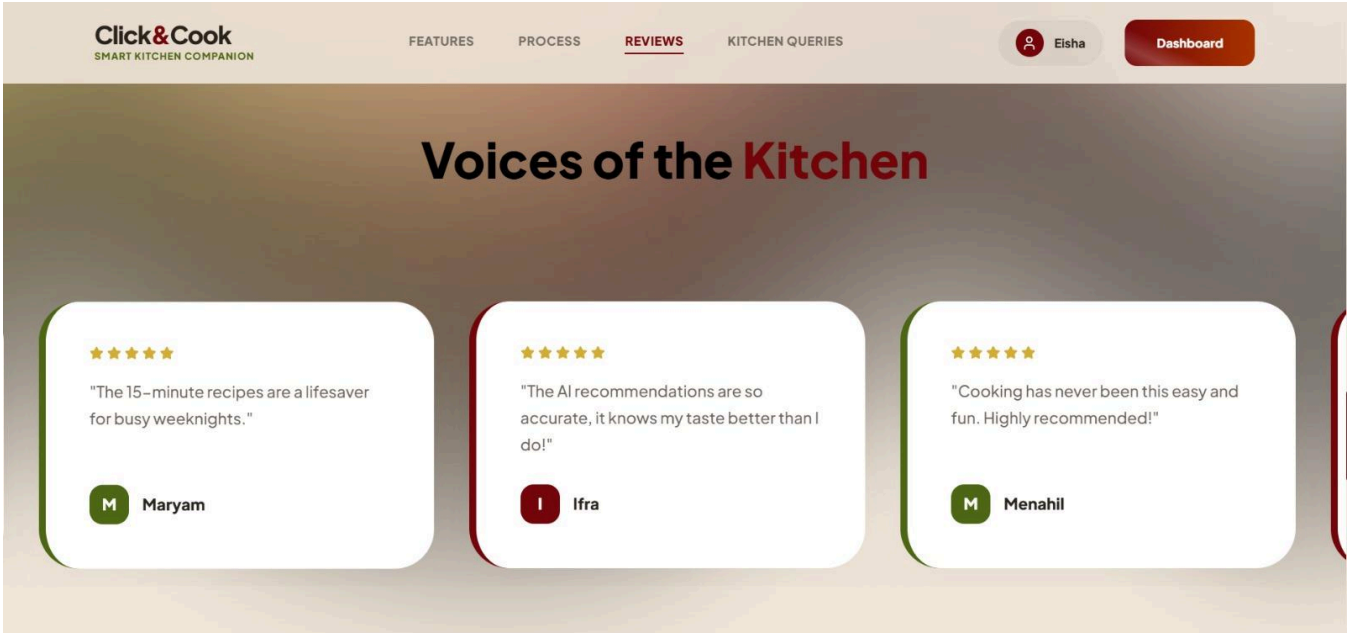
- Meal Planning: Weekly meal calendar with nutritional totals and shopping list generation
- Social Features: Share recipes publicly, follow other users, recipe collections
- Voice Interface: Voice-commanded recipe generation using Web Speech API
- Ingredient Scanner: Camera-based ingredient detection using computer vision
- Nutritional Tracking: Daily calorie and macro tracking dashboard
- Mobile Application: React Native mobile app for iOS and Android
- Recipe Ratings: Community rating system for AI-generated recipes
- Multi-language Support: Recipe generation in Urdu and other languages
- Premium AI Models: Integration with GPT-4 or Claude for enhanced recipe quality
- Grocery Integration: Direct ingredient ordering through grocery delivery APIs

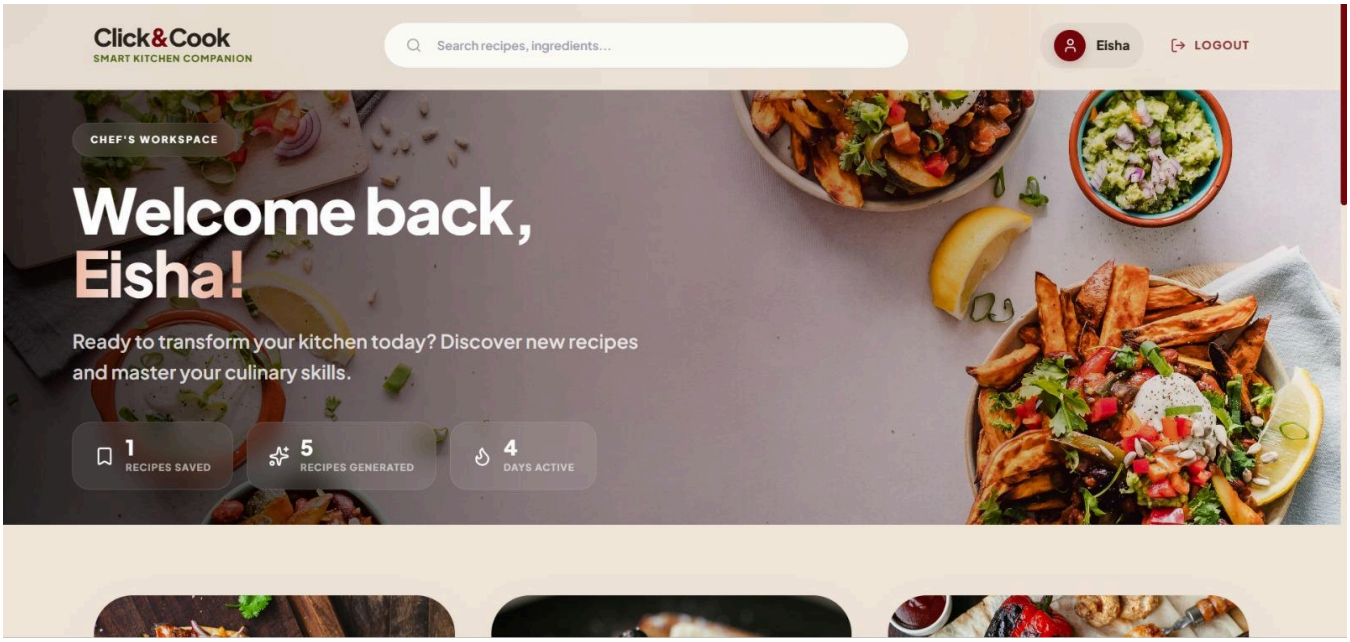
14. Images

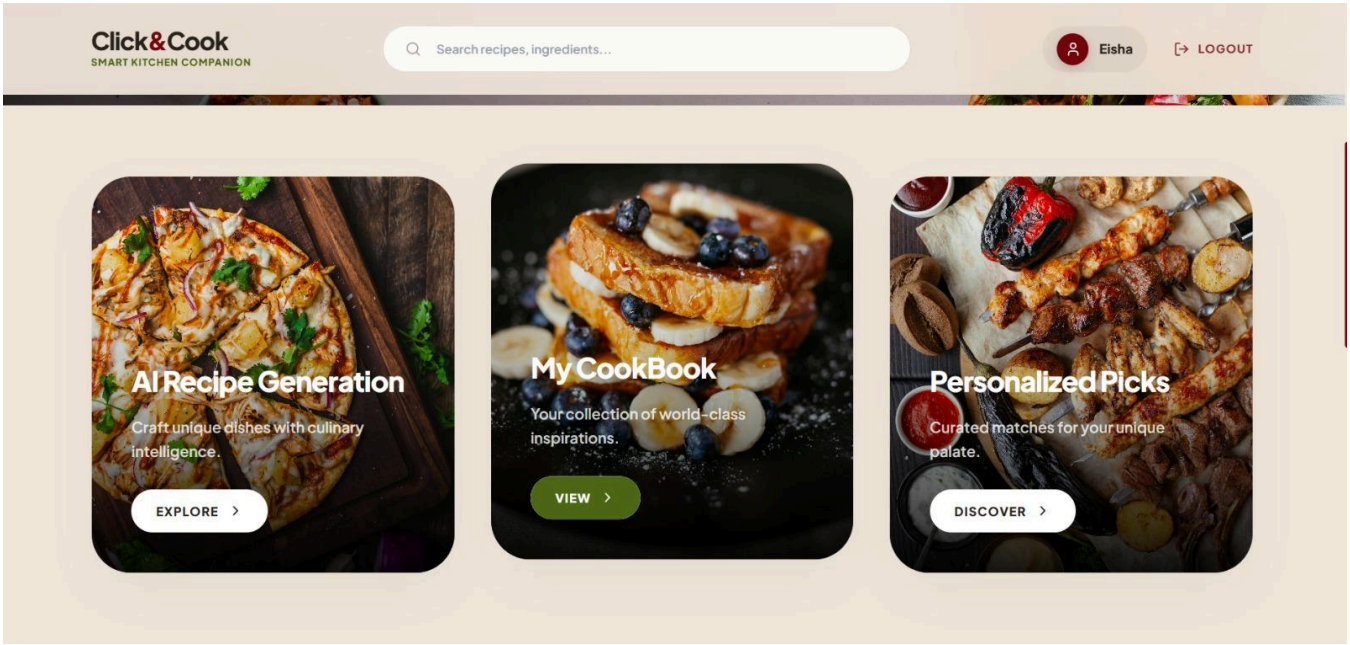

Click&Cook
SMART KITCHEN COMPANION
FEATURES
PROCESS
REVIEWS
KITCHEN QUERIES
Login
Get Started
THE FUTURE OF HOME COOKING
Cook Smarter, Eat Better Every Day
Transform your kitchen experience with AI-powered recipes, smart meal planning, and a community of food lovers. Your perfect meal is just one click away.
Start Cooking Now
QUICK PREP
15 Mins

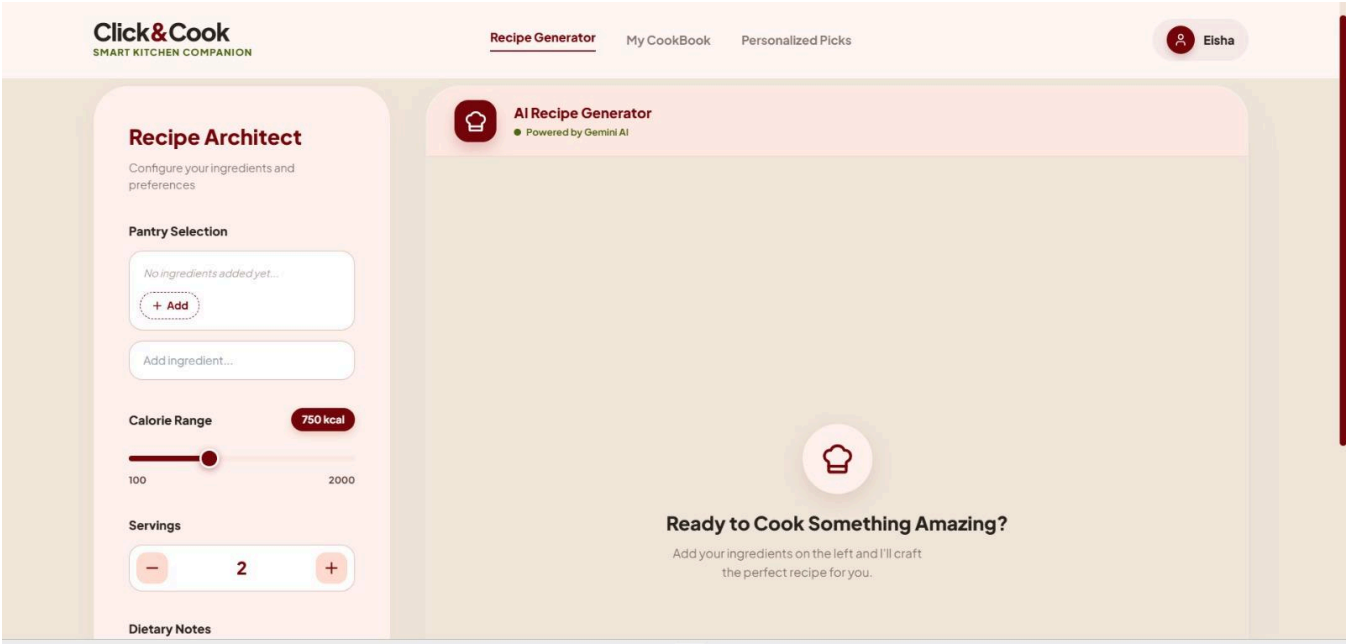

Click&Cook
SMART KITCHEN COMPANION
FEATURES
PROCESS
REVIEWS
KITCHEN QUERIES
Login
Get Started
Kitchen Superpowers
Everything you need to become a master chef in your own home.
AI Recipe Generation
Just tell us what's in your fridge, and our AI will craft a gourmet recipe in seconds. No waste, just great taste.
Save Your Favorites
Build your personal digital cookbook and access your recipes from any device, anywhere.
+2k

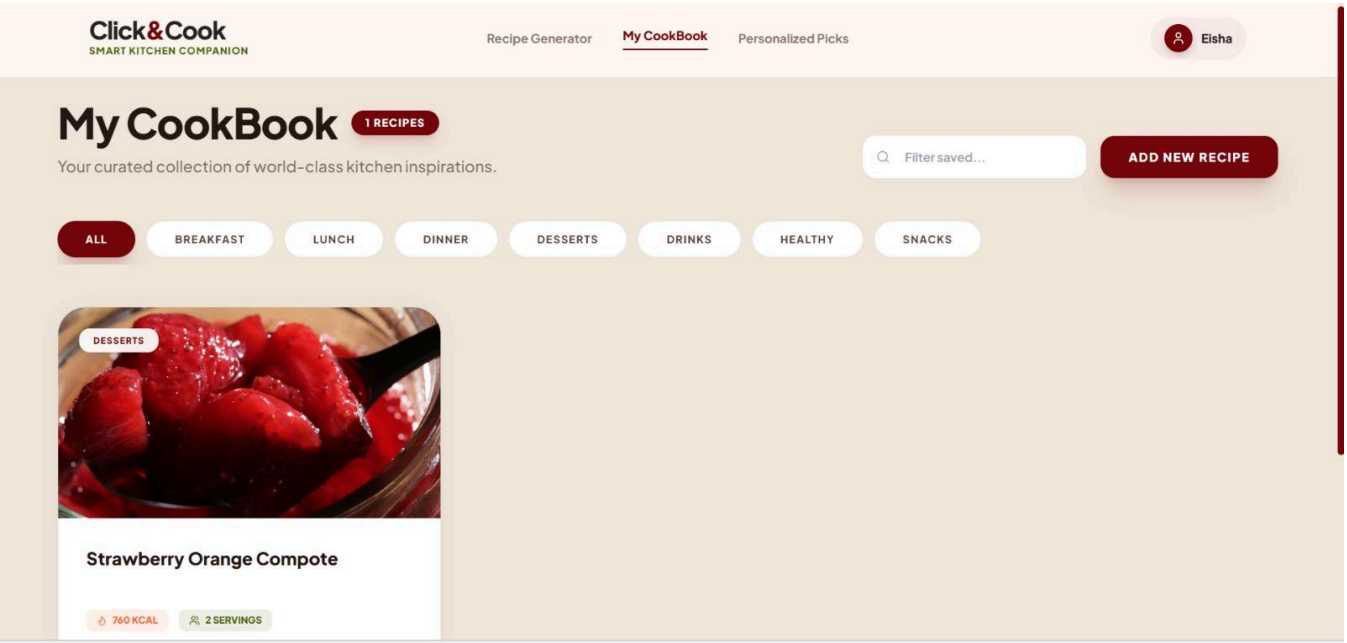

Click&Cook
SMART KITCHEN COMPANION
FEATURES
PROCESS
REVIEWS
KITCHEN QUERIES
Eisha
Dashboard
The Journey to Great Food
1
Input Your Pantry
Tell us what ingredients you have on hand or what you're craving.
Get AI Inspiration
Instantly receive creative, step-by-step recipes tailored to your skill level.
3
Cook & Enjoy
Follow the interactive guide and plate your masterpiece. Bon appétit!

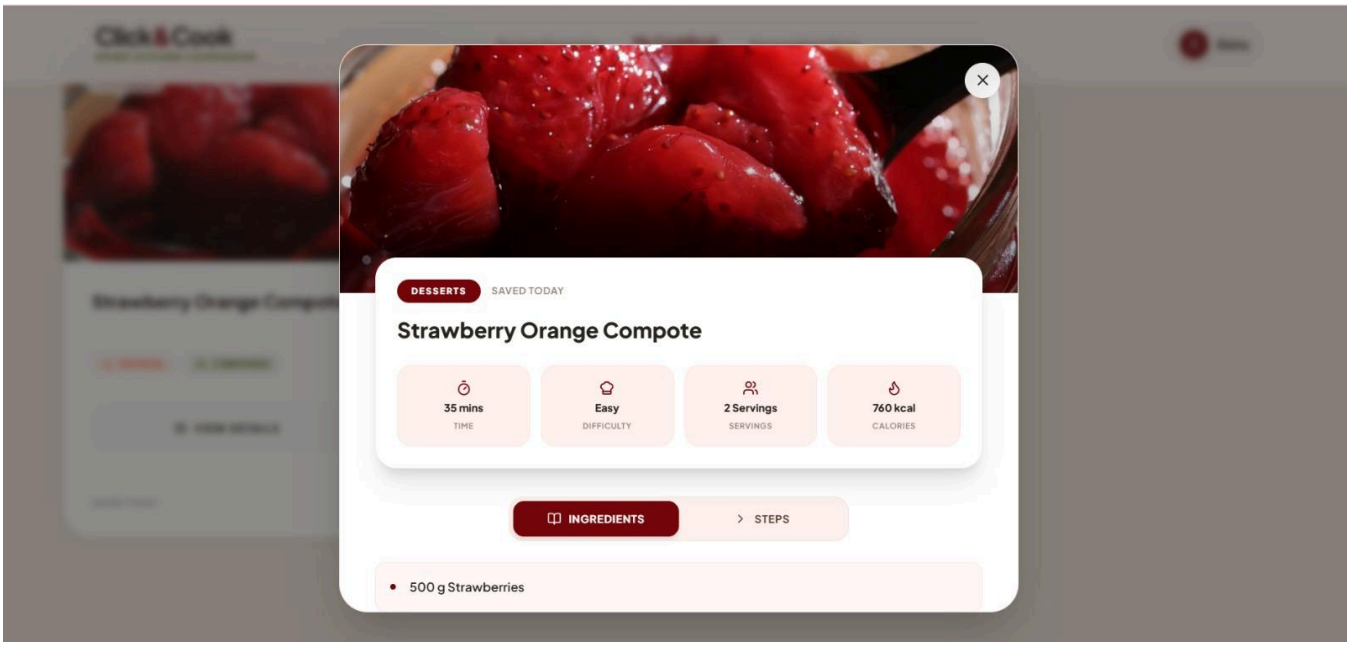

Click&Cook
SMART KITCHEN COMPANION
FEATURES
PROCESS
REVIEWS
KITCHEN QUERIES
Eisha
Dashboard
Voices of the Kitchen
"The 15-minute recipes are a lifesaver for busy weeknights."
M
Maryam
"The AI recommendations are so accurate, it knows my taste better than I do!"
I
Ifra
"Cooking has never been this easy and fun. Highly recommended!"
M
Menahil

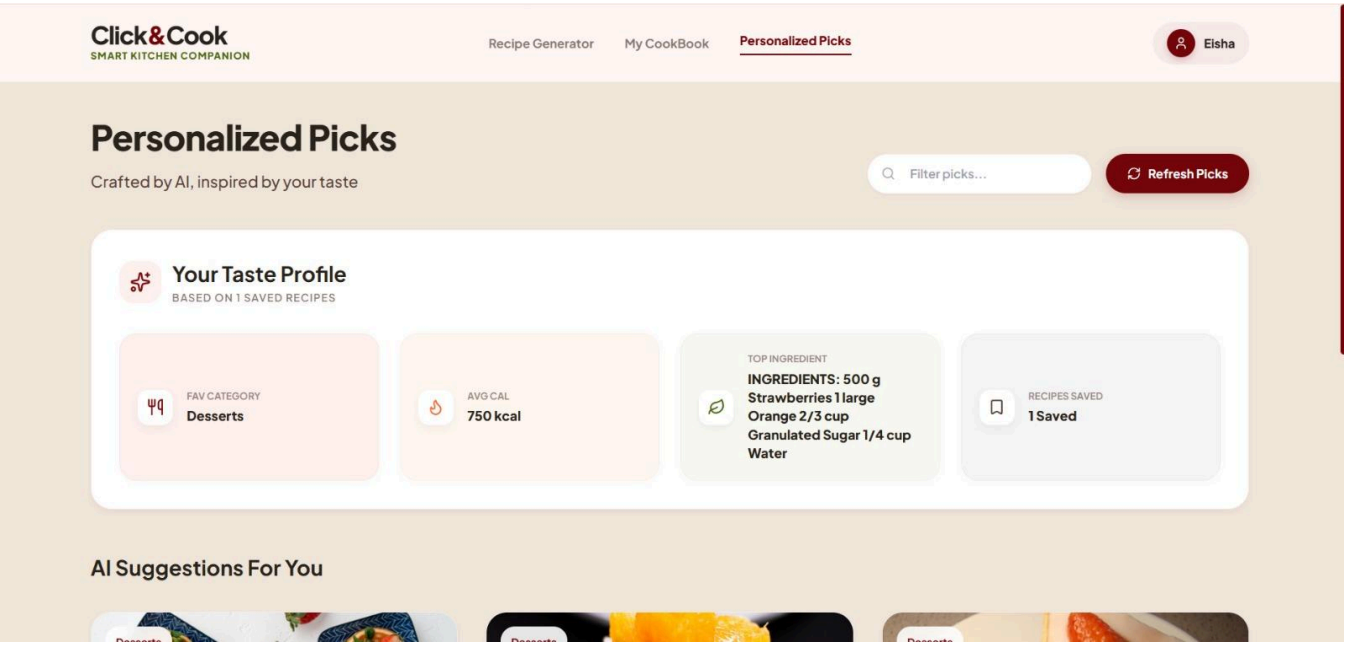

Click&Cook
SMART KITCHEN COMPANION
Search recipes, ingredients...
Eisha
LOGOUT
CHEF'S WORKSPACE
Welcome back,
Eisha!
Ready to transform your kitchen today? Discover new recipes and master your culinary skills.
1
RECIPES SAVED
5
RECIPES GENERATED
4
DAYS ACTIVE

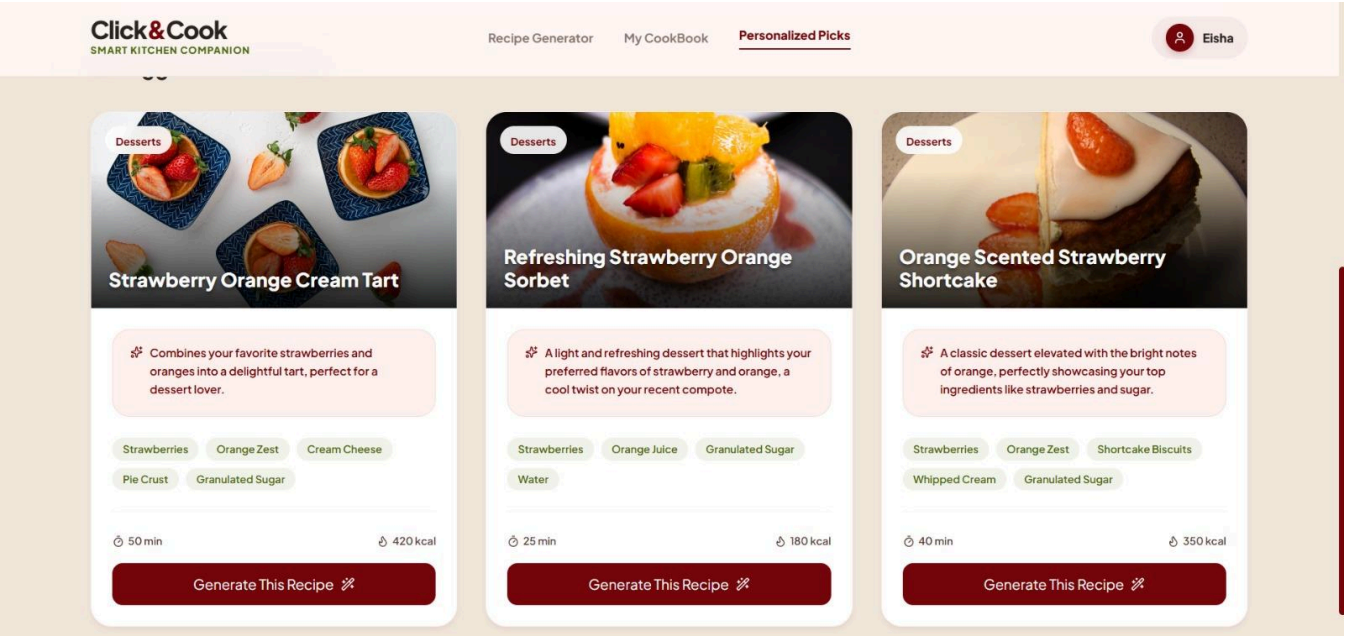

Click&Cook
SMART KITCHEN COMPANION
Search recipes, ingredients...
Eisha
LOGOUT
AI Recipe Generation
Craft unique dishes with culinary intelligence.
EXPLORE
My CookBook
Your collection of world-class inspirations.
VIEW
Personalized Picks
Curated matches for your unique palate.
DISCOVER

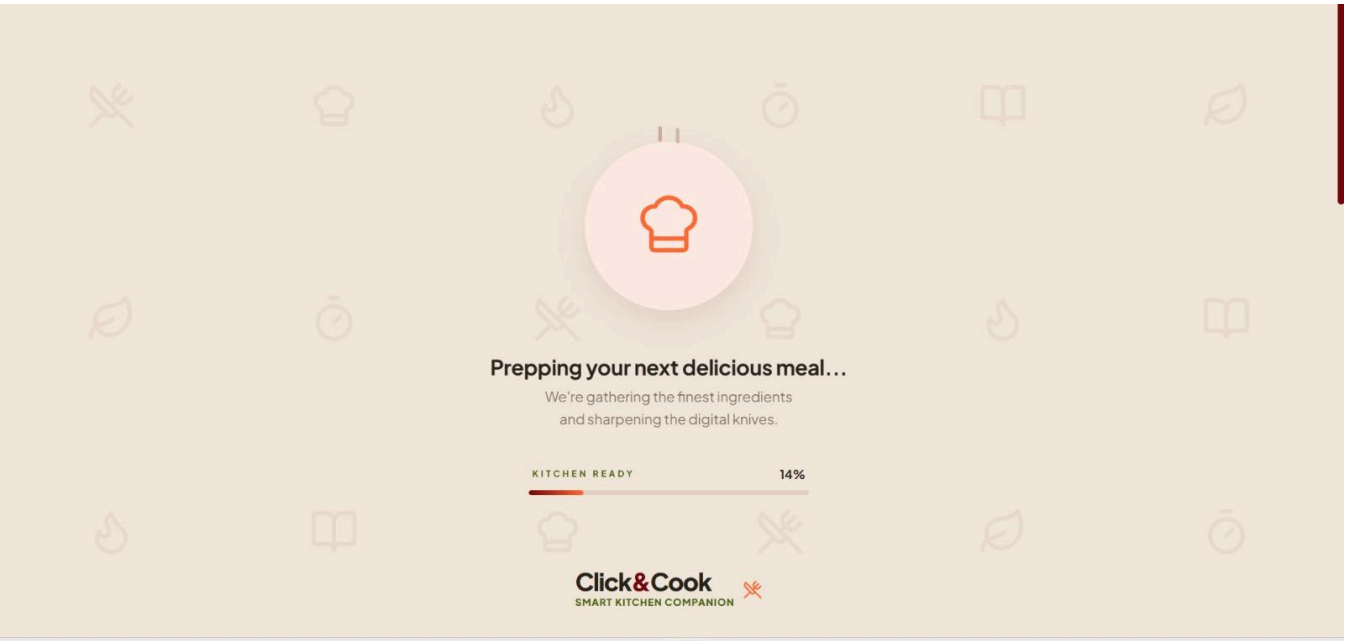

Click&Cook
SMART KITCHEN COMPANION
Recipe Generator
My CookBook
Personalized Picks
Eisha
Recipe Architect
Configure your ingredients and preferences
Pantry Selection
No ingredients added yet...
+ Add
Add ingredient...
Calorie Range
750 kcal
100
2000
Servings
2
Dietary Notes
AI Recipe Generator
Powered by Gemini AI
Ready to Cook Something Amazing?
Add your ingredients on the left and I'll craft the perfect recipe for you.

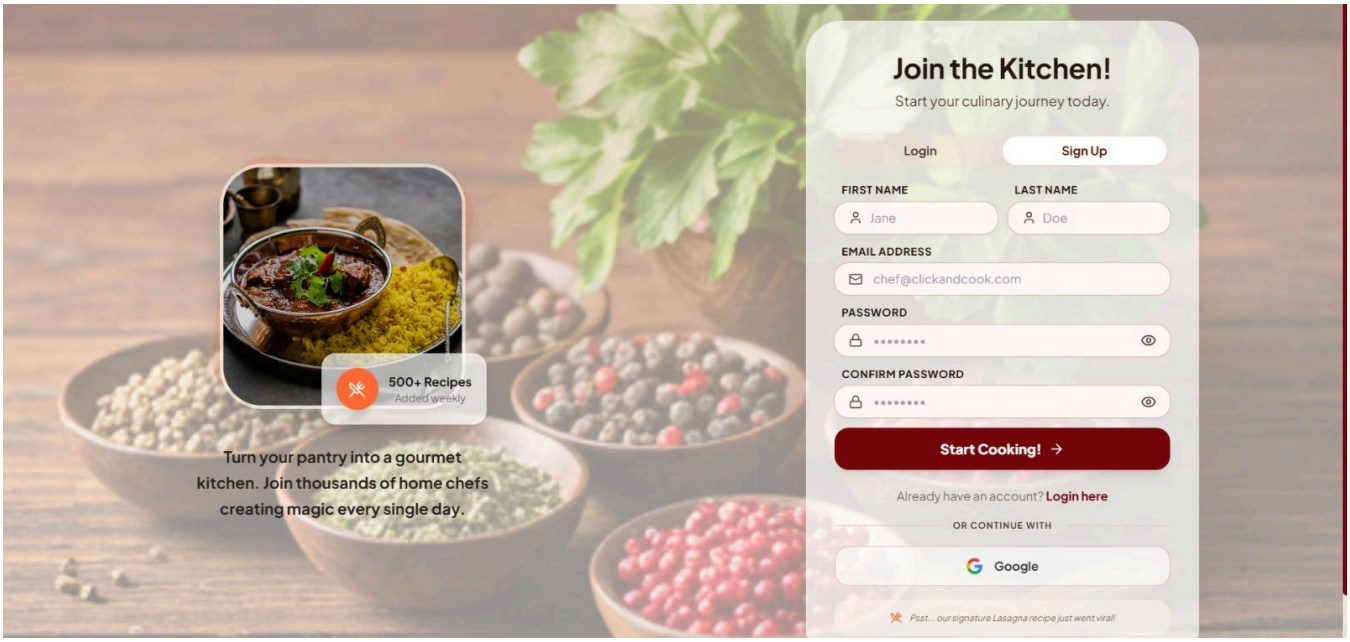

Click&Cook
SMART KITCHEN COMPANION
Recipe Generator
My CookBook
Personalized Picks
Eisha
My CookBook
1 RECIPES
Your curated collection of world-class kitchen inspirations.
Filter saved...
ADD NEW RECIPE
ALL
BREAKFAST
LUNCH
DINNER
DESSERTS
DRINKS
HEALTHY
SNACKS
DESSERTS
Strawberry Orange Compote
760 KCAL
2 SERVINGS


DESSERTS
SAVED TODAY
Strawberry Orange Compote
35 mins
TIME
Easy
DIFFICULTY
2 Servings
SERVINGS
760 kcal
CALORIES
INGREDIENTS
STEPS
500 g Strawberries


Click&Cook
SMART KITCHEN COMPANION
Recipe Generator
My CookBook
Personalized Picks
Elisha
Personalized Picks
Crafted by AI, inspired by your taste
Filter picks...
Refresh Picks
Your Taste Profile
BASED ON 1 SAVED RECIPES
FAV CATEGORY
Desserts
AVG CAL
750 kcal
TOP INGREDIENT
INGREDIENTS: 500 g Strawberries 1 large Orange 2/3 cup Granulated Sugar 1/4 cup Water
RECIPES SAVED
1 Saved
AI Suggestions For You


Click&Cook
SMART KITCHEN COMPANION
Recipe Generator
My CookBook
Personalized Picks
Eisha
Desserts
Strawberry Orange Cream Tart
Combines your favorite strawberries and oranges into a delightful tart, perfect for a dessert lover.
Strawberries
Orange Zest
Cream Cheese
Pie Crust
Granulated Sugar
50 min
420 kcal
Generate This Recipe
Desserts
Refreshing Strawberry Orange Sorbet
A light and refreshing dessert that highlights your preferred flavors of strawberry and orange, a cool twist on your recent compote.
Strawberries
Orange Juice
Granulated Sugar
Water
25 min
180 kcal
Generate This Recipe
Desserts
Orange Scented Strawberry Shortcake
A classic dessert elevated with the bright notes of orange, perfectly showcasing your top ingredients like strawberries and sugar.
Strawberries
Orange Zest
Shortcake Biscuits
Whipped Cream
Granulated Sugar
40 min
350 kcal
Generate This Recipe


Prepping your next delicious meal...
We're gathering the finest ingredients
and sharpening the digital knives.
KITCHEN READY
14%
Click&Cook
SMART KITCHEN COMPANION


500+ Recipes
Added weekly
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Sign Up
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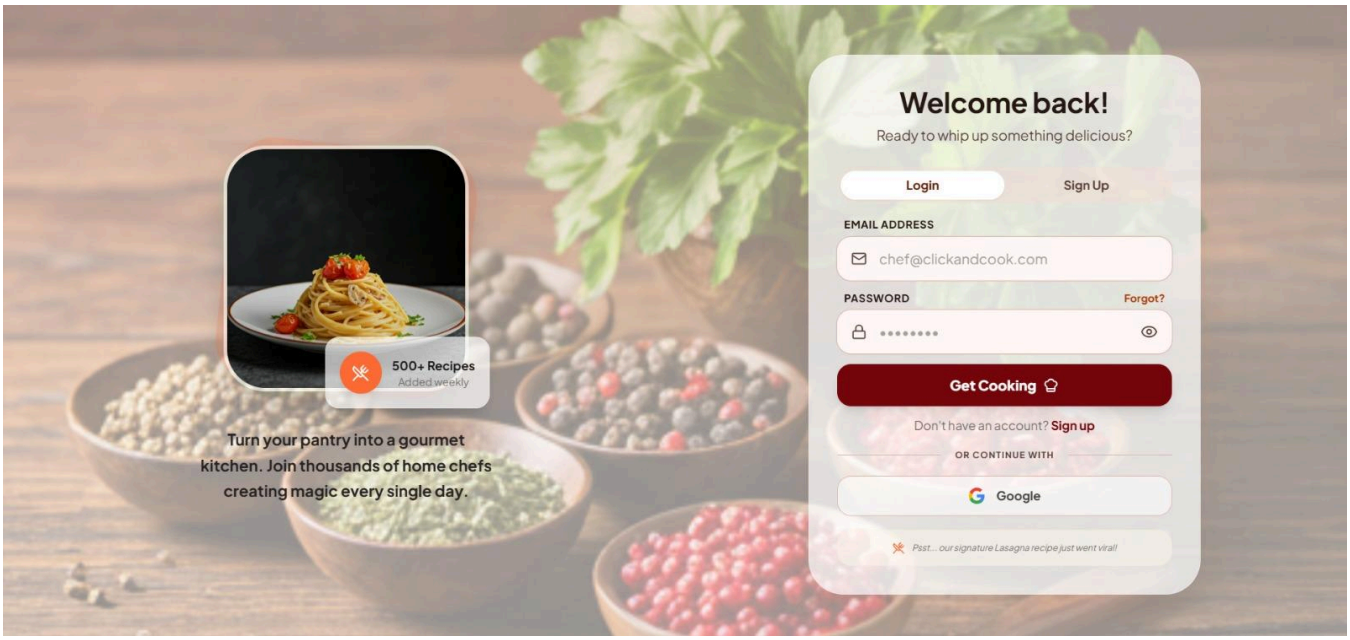
Jane
LAST NAME
Doe
EMAIL ADDRESS
chef@clickandcook.com
PASSWORD
CONFIRM PASSWORD
Start Cooking!
Already have an account? Login here
OR CONTINUE WITH
Google
Psst... our signature Lasagna recipe just went viral!


500+ Recipes
Added weekly
Turn your pantry into a gourmet kitchen. Join thousands of home chefs creating magic every single day.
Welcome back!
Ready to whip up something delicious?
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Get Cooking
Don't have an account? Sign up
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Psst... our signature Lasagna recipe just went viral!

15. Conclusion

Click & Cook successfully demonstrates the powerful convergence of modern web technologies, artificial intelligence, and thoughtful user experience design. The application solves a real-world problem faced by home cooks daily — transforming available ingredients into restaurant-quality meals with zero effort.

Built entirely from scratch using industry-standard technologies including React.js, Node.js, PostgreSQL, and Google's Gemini AI, the project showcases comprehensive full-stack development skills. The implementation covers the complete spectrum from database design and RESTful API development to dynamic AI integration and premium frontend engineering.

Key technical achievements include: seamless Google OAuth integration, automatic recipe categorization through AI response analysis, real-time food image fetching via Unsplash API, robust error handling with API key fallback mechanisms, and a sophisticated personalization engine that learns from user behavior.

The application is successfully deployed on production infrastructure (Vercel + Render + Neon) and is publicly accessible, demonstrating real-world deployment capabilities. This project represents not just a functional web application, but a comprehensive demonstration of modern software engineering principles applied to a meaningful, user-centric problem.